Household Fuel Price Risk and the Uneven Protective Value of Domestic Net Zero: Evidence from Westminster, London

May 25, 2026

Abstract

Fuel poverty identifies current affordability hardship, but it does not by itself measure future household exposure to fossil-fuel price shocks. This paper develops the concept of household fuel price risk and applies it to Westminster, London, using property-level or property-archetype modelling. The empirical strategy estimates required energy demand, baseline energy burden, price-shock sensitivity, and the distributional effect of domestic low-carbon interventions. The central argument is that domestic net zero is not automatically a household risk-protection strategy. Retrofit, clean heat, solar PV, battery storage, smart tariffs, and flexibility reduce unequal exposure only when housing fabric, heating systems, tariff design, market reform, tenure, and local governance align.

Keywords: fuel poverty; energy burden; price risk; domestic retrofit; net zero; Westminster; energy justice

1 Introduction

This paper reframes household energy inequality from a static problem of current fuel poverty into a dynamic problem of uneven exposure to future fossil-fuel price volatility. Westminster is an analytically useful case because high average wealth can obscure fuel-price vulnerability among households facing high housing costs, inefficient dwellings, private rental constraints, leasehold governance, heritage restrictions, and limited access to low-carbon assets.

The paper's starting point is therefore not that domestic net zero measures are inherently protective. A dwelling can become lower carbon without becoming less exposed to household price risk if the intervention leaves required heat demand high, shifts the household onto a more expensive fuel, excludes the occupant from favourable tariffs, or depends on building-level decisions outside the household's control. Conversely, the same intervention may be protective where fabric improvement, heating-system efficiency, tariff design, finance, and governance align. The key distributional question is who receives this protection and who remains exposed.

The central research question is: to what extent, and under what housing, market, and governance conditions, do domestic net zero interventions reduce unequal household exposure to fossil-fuel price volatility in Westminster?

The paper addresses three linked questions:

1. Which household-dwelling types face the largest required energy burden increase under fuel price shocks?

2. How do income, housing fabric, heating system, tenure, and vulnerability jointly produce fuel price risk?

3. How much risk reduction do retrofit, clean heat, PV, battery storage, smart tariffs, and integrated packages deliver, and for whom?

The contribution is to evaluate domestic net zero interventions as conditional household de-risking tools rather than as automatic bill-reduction devices. Empirically, the paper first establishes a current high-burden exposure layer using SAP-derived required energy costs and local income distributions. This layer identifies who is currently most exposed under baseline prices and why that exposure arises. It is then used as the foundation for future price-shock and intervention counterfactuals, where the object of analysis becomes household fuel price risk rather than current fuel poverty alone.

2 Conceptual Framework

The paper separates three concepts that are often conflated. Fuel poverty identifies present affordability hardship under current prices and required energy needs. Energy burden measures required or actual energy cost relative to residual household income. Household fuel price risk captures future exposure, sensitivity, and limited adaptive capacity under price-shock scenarios.

$$\text{Household fuel price risk} = \text{exposure} + \text{sensitivity} - \text{adaptive capacity}. \tag{1}$$

The implemented baseline burden indicators should therefore be read as current-price affordability measures, not as the final fuel-price-risk outcome. They provide the income–cost foundation for later shock scenarios, but they do not themselves represent future exposure unless fuel-price, tariff, and intervention counterfactuals are introduced.

In the empirical work, this distinction is operationalised through two related but non-equivalent burden measures. The first is a deterministic baseline required-energy-burden ratio: SAP-derived required energy cost divided by MSOA after-housing-cost disposable income. The second is a probabilistic threshold measure: the probability that a household drawn from the property's LSOA income distribution would face a required energy burden above 10 percent. The former is intuitive as a cost-to-income ratio; the latter is better suited to identifying local income-tail exposure. Neither measure is an official fuel-poverty classification, and neither should be read as future household fuel price risk before price shocks are modelled.

This matters for interpretation. A top-decile modelled burden-risk group identifies the high end of Westminster's current required-cost and income-distribution assumptions. It does not mean that every property in that group is occupied by a verified fuel-poor household, nor that all other properties are protected. Instead, it marks where required energy demand, local income sensitivity, and dwelling characteristics make high burden more probable under current prices. The future-facing risk analysis must then ask how this baseline exposure changes when gas prices, electricity prices, standing charges, tariff access, and policy mitigations are varied.

This distinction matters because decarbonisation is not the same as de-risking. Electrification can reduce emissions before it reduces household price risk where electricity prices remain linked to gas through wholesale market design. The protective value of heat pumps, PV, storage, smart tariffs, and flexibility therefore depends on the electricity-gas price ratio, tariff access, capital access, dwelling form, tenure, and governance.

3 Case and Data

The empirical case is Westminster, London. The preferred unit of analysis is the property or household-dwelling pair. Where household microdata are unavailable, the study uses EPC property records combined with LSOA/MSOA income and deprivation proxies to assign synthetic households or archetypes.

Westminster is not treated as a generic affluent borough. Its relevance lies in the coexistence of high aggregate wealth with residual-income pressure, older and heterogeneous housing, private renting, leasehold flats, conservation and heritage constraints, and social housing retrofit pathways. These features are not background descriptors; they shape exposure, sensitivity, and adaptive capacity. In a high-cost city, households can face fuel-price vulnerability through the combination of required heat demand and constrained residual income even where area-level averages appear comfortable.

The current enriched property master contains 124,980 properties across 126 LSOAs and 26 MSOAs. It combines EPC attributes, spatial identifiers, MSOA after-housing-cost income, LSOA income deprivation, official LSOA fuel poverty rates, and LSOA income-distribution outputs from the latest verified pipeline.

The latest EPC enrichment step links certificate records to the Westminster property master by UPRN, applies consistent UPRN normalisation in both merge and QA stages, and retains one selected EPC certificate per property using the latest lodgement or inspection date. The audited enriched stock preserves the master row count and reports zero unmatched master rows after the corrected linkage. EPC-derived tenure is used only as a proxy for tenure-related governance and adaptive capacity; the corrected analysis version classifies the stock as 58,744 private rented properties, 43,686 owner-occupied properties, 20,066 social rented properties, and 2,484 unknown-tenure properties.

The analysis uses required energy demand rather than only observed expenditure, because actual consumption may understate unmet need through underheating or self-rationing.

The resulting dataset is therefore best understood as a property-level modelling backbone for current required-burden exposure and future scenario analysis. It is not a direct observation of each household's income, consumption, or current tenure contract. This boundary is important: the modelling design is intended to reveal which property–income–governance combinations are likely to be exposed, not to certify household-level fuel-poverty status.

4 Methods

The empirical strategy is a property-level income–cost and scenario model. It links EPC-derived dwelling characteristics, synthetic LSOA income distributions, required energy costs, price-shock scenarios, domestic net zero intervention packages, and spatial-distributional outputs. The unit of analysis is property p in LSOA l and MSOA m. Where observed household incomes are unavailable, the model estimates the probability that a household drawn from the local residual-income distribution would face a given required-energy burden. This design is deliberately probabilistic: it does not claim to observe the income of the current occupant of each property.

4.1 Income–Cost Engine

The implemented income layer estimates one constrained lognormal after-housing-cost income distribution for each LSOA:

$$Y_l \sim \text{Lognormal}(\mu_l, \sigma_l^2), \tag{2}$$

where Y_l denotes annual residual household income. The distribution is parameterised by its mean $\bar{Y}_l$:

$$\mu_l = \ln(\bar{Y}_l) - \frac{1}{2}\sigma_l^2. \tag{3}$$

For each MSOA, the LSOA parameters are jointly estimated so that the property-weighted mean remains anchored to the MSOA after-housing-cost income, while the lower tail is differentiated by the LSOA income-deprivation signal. For a given regularisation parameter λ, the objective is:

$$\begin{aligned} \mathcal{L}_m(\lambda) = &\left[\ln\left(\frac{\sum_{l\in m} N_l \bar{Y}_l}{\sum_{l\in m} N_l}\right) - \ln(\bar{Y}_m)\right]^2 \\ &+ \lambda \sum_{l\in m} w_l \left[F_l(z) - d_l\right]^2, \end{aligned} \tag{4}$$

where N_l is the number of EPC property records in LSOA l, $w_l = N_l / \sum_{l \in m} N_l$, $\bar{Y}_m$ is MSOA residual income, $z = 19{,}009$ is the income threshold used in the current module, $F_l(\cdot)$ is the LSOA lognormal cumulative distribution function, and d_l is the LSOA income-deprivation score. The current implementation constrains $\sigma_l \in [0.10, 1.40]$ and searches a log-spaced grid of 100 λ values from 10^{-4} to 10^{-1}. Official LSOA fuel-poverty rates are used to select λ by minimising property-weighted RMSE, with weighted MAE and then the smaller λ as tie-breakers; they are not entered directly as the optimisation target.

Required energy cost is derived from SAP and floor area rather than from observed consumption, because actual expenditure may be suppressed by underheating or self-rationing. For property p, the SAP energy cost factor is:

$$\mathrm{ECF}_p = \begin{cases} (100 - \mathrm{SAP}_p)/16.21, & \mathrm{SAP}_p \geq 43.265, \\ 10^{(108.8 - \mathrm{SAP}_p)/120.5}, & \mathrm{SAP}_p < 43.265. \end{cases} \tag{5}$$

The baseline required annual energy cost is then:

$$C_{p,0} = \mathrm{ECF}_p \, \frac{A_p + 45}{0.2}, \tag{6}$$

where A_p is total floor area. Baseline required energy burden for a household with income Y_l is:

$$B_{p,0}(Y_l) = \frac{C_{p,0}}{Y_l}. \tag{7}$$

The current module therefore estimates burden-threshold probabilities:

$$P_{p,0}(\tau) = \Pr\left(B_{p,0}(Y_l) > \tau\right) = F_l\left(\frac{C_{p,0}}{\tau}\right). \tag{8}$$

The implemented baseline output uses $\tau = 0.10$. For modelled LSOA fuel-poverty comparison, properties with SAP grades A, B, or C are assigned a property poverty score of zero; other properties use $P_{p,0}(0.10)$. This score is a baseline affordability input, not a complete measure of household fuel price risk.

For robustness, the pipeline also reports a deterministic baseline burden ratio:

$$\widetilde{B}_{p,0} = \frac{C_{p,0}}{\bar{Y}_m}, \tag{9}$$

where $\bar{Y}_m$ is the MSOA after-housing-cost income point estimate. This measure is easier to interpret as a cost-to-income ratio, but it does not model within-LSOA income dispersion and is not calibrated to the local fuel-poverty comparison.

4.2 EPC Enrichment and High-Risk Pathways

The latest explanatory layer enriches the Westminster property master with EPC certificate records using UPRN-based linkage. UPRNs are standardised before both merging and QA checks, so that numeric and string representations of the same property identifier do not generate false unmatched cases. Where multiple EPC certificates are available for a property, the selected record is the latest `LODGEMENT_DATETIME`; where this is unavailable, the latest `INSPECTION_DATE` is used, and remaining ties fall back to the first available row. EPC fields are added with an `EPC_` prefix, the original property master is not overwritten, and the row count and row order of the master are preserved.

The enrichment is followed by an explicit audit stage covering unmatched master records, duplicate UPRNs, added-field missingness, EPC rating distributions, lodgement-year distributions, and original versus corrected tenure groups. The EPC-derived tenure variable is treated as a proxy for tenure-related adaptive capacity rather than as observed legal tenure. The analysis version, `EPC_TENURE_GROUP_FIXED`, classifies social rented records before private rented records to avoid misclassifying social rental descriptions as private rental cases.

The enriched stock is then transformed into a descriptive high-risk layer. Derived variables capture EPC performance, SAP-derived required energy cost, fabric weakness, electric heating exposure, gas connection, corrected tenure proxy, baseline required energy burden, and membership in modelled high-burden groups. The main current-condition groups are the top decile and top quintile of $P_{p,0}(0.10)$, denoted HR10 and HR20. Robustness groups use the top decile and top quintile of deterministic baseline required energy burden, denoted BR10 and BR20; an additional deterministic threshold group identifies properties with $\widetilde{B}_{p,0} > 0.10$. These groups describe current modelled burden probability and required-burden intensity; they are not official fuel-poverty classifications and do not by themselves constitute the future-facing household fuel price risk measure.

The distinction between HR and BR groups is analytically useful rather than merely technical. HR groups identify properties whose required costs interact with the local LSOA income distribution to produce a high probability of crossing a 10 percent burden threshold. BR groups identify properties with high deterministic required burden relative to the MSOA point estimate of after-housing-cost income. Their overlap is therefore expected to be partial: the two approaches should identify a related high-burden tail, while also showing whether income-distribution assumptions change who is treated as exposed.

Finally, high-risk properties are assigned to mutually exclusive descriptive pathways using priority rules. The current typology distinguishes inefficient large owner-occupied gas houses, social rented estate or block risk, private rented weak-fabric gas flats, electric-heated inefficient flats or maisonettes, high-demand inefficient maisonettes, and a residual high-risk group. The pathway typology is not a causal model. Its purpose is to provide an interpretable bridge between property-level burden modelling and the later price-shock and intervention simulations, so that de-risking outcomes can be reported by mechanism and governance pathway rather than only for the aggregate stock.

4.3 Price-Shock Risk Metrics

The price-shock layer converts the baseline income–cost engine into forward-looking risk metrics. Let $s \in S$ denote a fuel-price scenario, including gas-price, electricity-price, combined-fuel, standing-charge, and policy-mitigation cases. Required cost under scenario s is written as:

$$C_{p,s} = \pi_s^g Q_p^g + \pi_s^e Q_p^e + SC_s^g I_p^g + SC_s^e I_p^e - M_{p,s}, \tag{10}$$

where Q_p^g and Q_p^e are required gas and electricity quantities or archetype-imputed equivalents, π_s^g and π_s^e are scenario fuel prices, SC_s^g and SC_s^e are standing charges, I_p^g and I_p^e indicate fuel connections, and $M_{p,s}$ captures bill support or other mitigation. The scenario burden is:

$$B_{p,s}(Y_l) = \frac{C_{p,s}}{Y_l}, \tag{11}$$

and the burden uplift is:

$$\Delta B_{p,s}(Y_l) = B_{p,s}(Y_l) - B_{p,0}(Y_l). \tag{12}$$

The main risk indicators are defined as distributional rather than average outcomes. The scenario-specific threshold-crossing probability is:

$$P_{p,s}(\tau) = F_l\left(\frac{C_{p,s}}{\tau}\right). \tag{13}$$

For scenario weights ω_s, the expected threshold-crossing risk is:

$$R_p(\tau) = \sum_{s \in S} \omega_s P_{p,s}(\tau). \tag{14}$$

An affordability gap-at-risk can be expressed as the expected shortfall above an acceptable burden threshold:

$$G_{p,s}(\tau) = \mathrm{E}_{Y_l}\left[\max\{C_{p,s} - \tau Y_l, 0\}\right]. \tag{15}$$

Household Fuel Price-at-Risk is defined as a high quantile of the scenario burden distribution:

$$\mathrm{HFPaR}_{p,\alpha} = \inf\left\{b : \sum_{s\in S} \omega_s \mathbf{1}\left(B_{p,s} \leq b\right) \geq \alpha\right\}, \tag{16}$$

where α may be set to a policy-relevant upper-tail level such as 0.90 or 0.95. These metrics keep fuel poverty, energy burden, and fuel price risk analytically separate: the first concerns current hardship, the second is a cost-to-income ratio, and the third measures future exposure and sensitivity under price shocks.

4.4 Intervention Counterfactuals

Domestic net zero interventions are represented as counterfactual changes to required demand, fuel structure, tariff exposure, and mitigation access. Let k denote an intervention package, such as fabric retrofit, clean heat, PV, battery storage, smart tariffs, or an integrated package. The post-intervention scenario cost is:

$$C_{p,s,k} = a_{p,k}\widetilde{C}_{p,s,k} + (1 - a_{p,k})C_{p,s}, \tag{17}$$

where $a_{p,k} \in [0, 1]$ represents technical, tenure, leasehold, heritage, roof, finance, or tariff feasibility. This term is central to the paper's interpretation: an intervention produces household risk protection only where access and market conditions allow the technical change to translate into lower exposure.

The intervention-specific threshold risk and Fuel Price-at-Risk are:

$$P_{p,s,k}(\tau) = F_l\left(\frac{C_{p,s,k}}{\tau}\right), \qquad \mathrm{HFPaR}_{p,\alpha,k} = Q_\alpha\left(\{B_{p,s,k}\}_{s\in S}\right). \tag{18}$$

The de-risking dividend is the reduction in upper-tail burden risk:

$$D_{p,\alpha,k} = \mathrm{HFPaR}_{p,\alpha,0} - \mathrm{HFPaR}_{p,\alpha,k}. \tag{19}$$

Results will be stratified by income distribution, EPC rating, dwelling form, heating system, tenure, leasehold and heritage constraints, and LSOA/MSOA location. The empirical question is therefore not whether a low-carbon measure reduces emissions in engineering terms, but who receives measurable price-risk protection once required demand, tariffs, fuel-market pass-through, and local governance constraints are jointly considered.

5 Expected Outputs

The empirical outputs are designed to move from an implemented income–cost baseline to forward-looking price-risk and intervention comparisons. The latest verified modelling round already provides the first layer of this sequence: a property-level baseline required-energy-burden engine for Westminster, built from constrained LSOA income distributions and SAP-derived required energy costs. The run labelled `20260520_213323` covers 124,980 EPC property records across 126 LSOAs and 26 MSOAs. It preserves MSOA after-housing-cost income consistency while allowing LSOA differentiation mainly through income dispersion and lower-tail probability, rather than by claiming directly observed household incomes.

5.1 Implemented Baseline Outputs

The current outputs support three baseline products. First, the model estimates LSOA lognormal income parameters, including μ_l, σ_l, lower-tail probabilities below £19,009, and the selected MSOA-level regularisation parameter λ. In the latest verified run, the ratio of LSOA mean income to MSOA income is tightly anchored, ranging from 0.999650 to 1.000309, while σ_l ranges from 0.4192 to 1.4000. Only 3 of 126 LSOAs sit at the new upper bound, compared with 31 of 126 under the previous bound of 0.9. This indicates that within-MSOA differentiation is being carried primarily by dispersion and tail probability.

Second, the model produces property-level SAP-derived required energy cost and the probability that required energy burden exceeds a 10 percent threshold. This output should be interpreted as baseline required-energy affordability exposure, not as a complete household fuel price risk measure. It provides the denominator and cost foundation for later shock scenarios, while avoiding dependence on actual expenditure that may be suppressed by underheating or self-rationing.

Third, the baseline module generates a modelled LSOA fuel-poverty comparison used for calibration diagnostics. The latest run has a mean modelled-minus-official LSOA fuel-poverty-rate error of -0.0174, a median error of -0.0270, and a worst absolute error of 0.133. This residual gap is substantively useful rather than merely technical: it shows that income-tail matching alone does not reproduce official fuel-poverty geography, reinforcing the paper's distinction between income deprivation, required energy burden, dwelling fabric, heating systems, tenure, and future price risk.

5.2 Implemented High-Risk Explanation Outputs

The latest high-risk explanation stage adds an audited EPC-enriched stock and a descriptive pathway typology. The enriched master retains 124,980 Westminster property records. After corrected UPRN standardisation, the QA workbook reports a complete match between the master properties and EPC certificate records, with zero unmatched master rows. The corrected EPC tenure proxy distributes the stock into 58,744 private rented properties, 43,686 owner-occupied properties, 20,066 social rented properties, and 2,484 unknown-tenure properties. These tenure categories are analytical proxies and should be interpreted as adaptive-capacity and governance indicators rather than as verified household tenure records.

The implemented high-risk groups remain exploratory and model-based. HR10 contains the 12,500 properties in the top decile of the probability that required energy burden exceeds 10 percent; HR20 contains the 24,996 properties in the top quintile. The deterministic burden robustness groups contain 12,498 BR10 properties and 24,996 BR20 properties, while only 737 properties have a baseline required energy burden above 10 percent under the deterministic baseline measure. HR10 and BR10 overlap for 7,128 properties, or 57.0 percent of each group. This overlap is large enough to show that the probabilistic and deterministic indicators identify a related high-burden tail, but incomplete enough to justify reporting both.

The strongest descriptive signal in the current outputs is the interaction between dwelling size and energy performance. EPC E–F–G properties above 120 m^2 have an HR10 share of 69.68 percent and an HR20 share of 85.21 percent, with a median baseline burden of 7.26 percent. E–F–G houses also show high within-type exposure, with an HR10 share of 56.08 percent. By contrast, E–F–G flats have a lower within-type HR10 share of 19.38 percent, but remain important because flats are numerous in the Westminster stock.

The pathway table should be read as a targeting and interpretation device, not as a final causal attribution model. It separates two policy-relevant rankings: the highest within-pathway risk is concentrated in inefficient large owner-occupied gas houses, high-demand inefficient maisonettes, and social rented estate/block risk, while the largest contributions to all HR10 properties come from social rented estate/block risk, private rented weak-fabric gas flats, electric-

Table 1: Current high-risk pathway typology from the latest explanatory run

Pathway	Properties	Stock	HR10	HR10 contrib.
Inefficient large owner-occupied gas houses	2,802	2.24%	50.25%	11.26%
Social rented estate/block risk	7,016	5.61%	22.66%	12.72%
Private rented weak-fabric gas flats	34,672	27.74%	4.18%	11.60%
Electric-heated inefficient flats/maisonettes	10,256	8.21%	13.83%	11.34%
High-demand inefficient maisonettes	2,220	1.78%	25.81%	4.58%
Other high-risk residual group	13,063	10.45%	46.40%	48.49%
Not high risk / comparison stock	54,951	43.97%	0.00%	0.00%

heated inefficient flats/maisonettes, and the residual high-risk group. The next empirical outputs should therefore report both risk intensity and stock-scale contribution.

The residual group is especially important for the next stage. It is too large to remain a black-box category once the paper moves from current exposure into price-shock risk. The planned analysis should separate income-driven residual cases, spatial-deprivation concentrations, mixed housing-and-income mechanisms, and unusual dwelling profiles. This will help distinguish households for whom technical retrofit is the central risk-reduction route from those for whom social tariffs, standing-charge reform, income support, or local assistance may be more relevant.

5.3 Risk and Intervention Outputs

The next empirical layer will convert the baseline engine into price-shock outputs. These will include LSOA/MSOA maps, property-archetype summaries, distributional plots, and threshold-crossing estimates under gas-price, electricity-price, combined-fuel, standing-charge, and policy-mitigation scenarios. The paper will report not only average burden changes, but also upper-tail risk, burden uplift, and the probability that households cross policy-relevant burden thresholds.

A central output is the de-risking dividend:

$$\text{De-risking dividend} = \text{Fuel Price-at-Risk}_{\text{before}} - \text{Fuel Price-at-Risk}_{\text{after}}. \tag{20}$$

This metric will be estimated by intervention package and household-dwelling condition, including fabric retrofit, clean heat, PV, battery storage, smart tariffs, and integrated packages. Its interpretation will remain conditional: an intervention produces household price-risk protection only where technical feasibility, tenure and leasehold governance, tariff access, electricity/gas price ratios, and market pass-through allow the engineering change to reduce household exposure.

5.4 Figures and Tables

Expected tables will summarise baseline burden probabilities, modelled fuel-poverty diagnostics, price-shock burden uplift, Fuel Price-at-Risk, and intervention-specific de-risking dividends by income group, EPC rating, dwelling form, heating system, tenure, and LSOA/MSOA. Expected figures will include: the conceptual distinction between fuel poverty, energy burden, and household fuel price risk; maps of baseline required-energy-burden exposure; distributional plots of shock-induced burden uplift; intervention de-risking comparisons; and maps of de-risking gaps.

The latest high-risk explanation stage also identifies near-term manuscript figures: a side-by-side comparison of HR10 share within each pathway and each pathway's contribution to all HR10 properties; an EPC-band by floor-area heatmap coloured by HR10 share; a tenure-proxy by EPC-band heatmap; an LSOA map of HR10 share; an LSOA map of the dominant HR10 pathway; and, if needed, an alluvial chart linking tenure, property type, EPC band, and pathway.

These should be promoted to `paper/figures/` only after specific figure files are selected and checked.

Before intervention results are reported, the spatial outputs should establish whether different mechanisms concentrate in different parts of Westminster. Priority maps include LSOA HR10 share, LSOA HR20 share, dominant HR10 pathway, social rented estate/block concentration, electric-heated inefficient flat concentration, and residual high-risk concentration. These maps would support the paper's claim that fuel price risk is produced by locally specific combinations of housing form, income sensitivity, and governance constraints rather than by a uniform borough-wide condition.

The synced LSOA income PDF panels from the latest run are useful writing and diagnostic references. They show the estimated lognormal income distributions under selected λ values, with poverty-line and MSOA-mean markers. If promoted into the manuscript, they should be captioned as synthetic constrained income distributions for burden modelling, not as observed household income distributions and not as direct household fuel price risk outputs.

The paper will ultimately identify de-risking gaps: places and household-dwelling types where fuel price protection is most needed but low-carbon intervention is least accessible. These gaps are expected to be especially important where Westminster's high housing costs, private renting, leasehold flats, heritage constraints, district-heating governance, and uneven tariff or asset access interrupt the translation from domestic net zero investment into household price-risk protection.

6 Discussion

The expected policy implication is that domestic net zero should be treated as a distributional household risk-protection agenda. Protection depends on the alignment of fabric retrofit, heating technology, distributed generation, flexibility, tariff design, electricity market reform, capital access, and tenure-sensitive governance.

In Westminster, this means paying particular attention to private renters, leasehold flats, HMOs, heritage-constrained dwellings, district-heating governance, social housing delivery pathways, and households whose vulnerability is masked by high neighbourhood average wealth.

The latest high-risk explanatory results sharpen this argument. Westminster's current modelled high-burden tail is not produced by a single deficit such as low EPC performance alone. It emerges through multiple routes: high required heat demand, weak fabric, income sensitivity, tenure governance, electricity price exposure, and limited household control over building-level interventions. This matters because the policy question is not simply where average bills are high, but where future price shocks are likely to meet constrained adaptive capacity.

The interaction between dwelling size and EPC performance is especially important. Inefficient large dwellings have the highest within-type modelled burden probability, showing that poor fabric becomes most consequential where required energy demand is high. Yet the pathway typology also shows that intensity and scale are different policy problems. Inefficient large owner-occupied gas houses are a small but high-intensity group, while private rented weak-fabric gas flats and electric-heated inefficient flats or maisonettes contribute materially to the high-risk tail because they are common in the stock. A targeting strategy based only on the highest within-group risk would miss large numbers of exposed households; a strategy based only on stock counts would miss the households facing the steepest required-burden pressure.

Tenure changes the route through which de-risking can occur. Owner occupation does not automatically imply adaptive capacity where heritage rules, planning constraints, capital cost, and disruption limit retrofit. Social rented estate and block risk is primarily a delivery and governance problem: tenants cannot individually decide the timing, specification, or financing of estate-level fabric works, heating-system changes, or metering reform. Private rented weak-fabric flats raise a different problem of split incentives, enforcement, and minimum-standard compliance, especially where moderate aggregate EPC ratings obscure component-level fabric weakness.

The electric-heating pathway reinforces the paper's caution about electrification. Electric-heated inefficient flats and maisonettes may already be exposed to high electricity unit costs. Unless fabric improvement, tariff reform, flexibility access, and electricity-market design reduce the household-facing price exposure, electrification can reduce emissions before it reduces household fuel price risk. The empirical strategy should therefore test clean heat, PV, batteries, smart tariffs, and integrated packages under explicit gas-to-electricity price-ratio assumptions rather than treating electrification as automatic protection.

The residual pathway is also substantively important. Nearly half of HR10 properties sit in the residual high-risk group, indicating that income sensitivity, local deprivation, and mixed mechanisms remain central to the current burden-risk distribution. The next modelling stage should therefore separate housing-driven, income-driven, and mixed-mechanism risk before estimating intervention de-risking dividends. This is not a weakness of the typology; it is evidence that the paper should avoid reducing household fuel price risk to a single housing-technology variable.

The broader implication is that domestic net zero policy must be evaluated against pathway-specific routes to protection. Fabric-first retrofit is most directly relevant where high required heat demand and poor energy performance dominate. Social housing pathways require block-level delivery, tenant protection, and post-intervention bill assessment. Private rented pathways require enforcement, landlord incentives, and tenant safeguards. Electric-heating pathways require tariff and flexibility reform as much as technology substitution. Residual income-driven pathways may need social tariffs, standing-charge reform, or income support before household risk can be materially reduced. The empirical test of domestic net zero is therefore not whether interventions look effective on average, but whether they reach the mechanisms through which exposure is produced.

7 Conclusion

Fuel poverty identifies current hardship; household fuel price risk identifies future exposure to price shocks. Net zero determines who receives protection only when domestic interventions are technically effective, financially accessible, and institutionally deliverable for the households most exposed.

The current Westminster evidence strengthens this framing. Modelled high-burden exposure is heterogeneous: large inefficient gas houses, social rented blocks, private rented weak-fabric flats, electric-heated inefficient flats and maisonettes, high-demand maisonettes, and residual income-sensitive cases represent different routes into risk and different routes out of it. A single technology narrative cannot explain this distribution.

The paper therefore asks not whether net zero reduces emissions, but whether, how, and for whom it credibly reduces household exposure to fossil-fuel price volatility. Its policy implication is correspondingly conditional: domestic net zero becomes household protection only where fabric, heating systems, tariffs, market design, finance, tenure, and local governance convert technical change into reduced upper-tail burden risk.